

Unit 7, Lesson 10: Multiplying Integers Using Integer Chips

Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

Learning Target

- ☐ I can use integer chips to multiply integers.

7.10.1 Warm-Up: Adding and Subtracting

1. Place a digit in each blue box so that the expressions in the first row have the same value and the expression in the second row has the largest value. Use only digits 1 to 6, at most one time each.

Description of the Values	Expressions	Digits to Choose From
Same value	— + — —	1 2 3 4 5 6
Largest value	— — —	

7.10.2 Exploration: Using Integer Chips to Represent Multiplication

1. The addition expression $7 + 7 + 7$ can be written as ____ $\cdot 7$.

This is because repeated addition of integers can be expressed as

- ☐ subtraction.
- ☐ multiplication.
- ☐ division.

2. The expression 5×2 can be interpreted as adding ____ groups of ____ or adding ____ groups of ____.

3. Use integer chips to represent one of these interpretations.

Recall that  = -1 and  = $+1$.

Draw a picture of your representation.

4. Describe how the integer chips represent the value of the expression 5×2 .
5. The expression $5 \times (-2)$ can be interpreted as adding ____ groups of ____ or adding ____ groups of ____.
6. Explain which interpretation of $5 \times (-2)$ you think would be easier to represent using integer chips.

7. Discuss your choice with your partner. Then, work together to use integer chips to represent one of the interpretations. Draw a picture of the representation.
8. Describe how the integer chips represent the value of the expression $5 \times (-2)$.
9. What is the value of the expression $5 \times (-2)$?



Commutative Property of Multiplication

$$a \times b = b \times a$$

The ***commutative property of multiplication*** states that the order in which multiplication is performed does not change the value of the expression.

10. Using the commutative property of multiplication, what is the value of the expression -2×5 ?

7.10.3 Collaboration: Are Zero Pairs Necessary for Multiplication?

Rashaad and Sebastián were discussing how to represent -2×5 using integer chips.

Sebastián said, “I don’t know how you can have negative 2 groups.”

Rashaad said, “I agree. We were able to represent 5×-2 . Why can’t we figure out how to represent the equivalent expression -2×5 ?”

Sebastián then asked, “When we added and subtracted using integer chips, we used zero pairs. Do you think using zero pairs would help?”

1. Discuss with your partner Sebastián’s question, “Do you think using zero pairs would help?”
2. Briefly explain your answer to his question. Draw integer chips as part of your explanation if that is helpful.

7.10.4 Exploration: Using Integer Chips to Represent Multiplication with Negative Groups

In the subtraction problem $5 - 2$, the -2 implies that 2 should be taken away from 5. In the expression -2×5 , the -2 can be interpreted as removing 2 groups of 5.

Use the steps below to discover how to represent -2×5 using integer chips.

Step 1: First, use your integer chips to create 2 groups of 5 zero pairs. Draw the 2 groups of 5 zero pairs in the box below.



Step 2: Recall that the expression -2×5 is interpreted as removing 2 groups of 5. From your integer chips, remove 2 groups of 5 positive integer chips. In the box above, show removing 2 groups of positive 5 integer chips by crossing them out.

The value of the expression $5 \times (-2)$ is -10 . Due to the commutative property of multiplication, -2×5 is expected to have the same value.

Does the number of chips that remain match the expected value of -2×5 ?

- ☐ Yes,
- ☐ No,

the value of -2×5 is _____ according to the model.

7.10.5 Try It: Multiplying Integers with Opposite Signs

1. Complete the table.

Expression	Visual Model Using Integer Chips	Product
$6 \times (-2)$		
$(4)(^{-}1)$		

2. Use the pattern of the multiplication in the table to complete the rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
 $\times$ 	

3. Multiplication is commutative, so ...

 $\times$  $=$ $\times$

7.10.6 Exploration: Multiplying a Negative Integer by a Negative Integer

1. Complete the statement.

The expression, $(-5)(-2)$, indicates that _____ groups
of _____ should be removed.

2. Use the steps below to discover how to represent $(-5)(-2)$ using integer chips.

Step 1: First, use your integer chips to create the set of zero pairs that will be needed to find $(-5)(-2)$. Draw the groups of zero pairs in the box below.



Step 2: Remove 5 groups of 2 negative integer chips.

Step 3: Remove any remaining zero pairs, if needed.

3. Complete the statement.

The value of expression $(-5)(-2)$ is _____.

4. With your partner, discuss whether or not you think the product when multiplying two negative values will always be positive. Explain your thinking.

7.10.7 Try It: Multiplying a Negative Integer by a Negative Integer

1. Complete the table.

Expression	Visual Model Using Integer Chips	Product
$-3 \times (-2)$		
$-1 \cdot -6$		

2. Complete each rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
 $\times$ 	
 $\times$ 	

7.10.8 Wrap-Up: Cool Down

1. Complete each rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
 × 	
 × 	
 × 	
 × 	